$$3V_C^2 = \frac{a}{P_C}$$

$$P_C = \frac{a}{3V_C^2} = \frac{a}{3(3b^2)} = \frac{a}{3 \times 9b^2} = \frac{a}{27\,b^2}$$

$$P_C = \frac{a}{27b^2} \quad \text{------ (6.32)}$$

substituting the values of V_c and P_c in equation (6.28),

$$3V_C = b + \frac{R\,T_C}{P}$$

$$3\,(3\,b) = b + \frac{R\,T_C}{\left(\frac{a}{27\,b^2}\right)}$$

$$9\,b - b = \left(\frac{R\,T_C}{a}\right) 27\,b^2$$

$$8\,b = \frac{T_C\,R\,27b^2}{a}$$

$$\therefore \quad T_C = \frac{8\,a\,b}{27\,R\,b^2} = \frac{8\,a}{27\,R\,b}$$

$$T_C = \frac{8\,a}{27\,R\,b} \quad \text{------ (6.33)}$$

The critical constants can be calculated using the values of van der waals constant of a gas and vice versa.

$$a = 3\,V_C^2\,P_C \quad \text{and} \quad b = \frac{V_C}{3}$$

6.7 Liquefaction of gases

For important commercial operations such as LPG and rocket fuels, we require gases in their liquid state. The liquefication methods are based on the Joule-Thomson effect. He observed appreciable cooling when the compressed gas is forced through an orifice plug into a low-pressure region. This phenomenon of lowering of temperature when a gas is made to expand adiabatically from a region of high pressure into a region of low pressure is known as Joule-Thomson effect. This effect is observed only below a certain temperature, which is a characteristic one for each gas. This temperature below which a gas obeys Joule-Thomson effect is called inversion temperature (T_i). This value is given using van der waals constants *a* and *b*.

$$T_i = \frac{2a}{Rb} \quad \text{--------- (6.34)}$$

Gases like O_2, He, N_2 and H_2 have very low T_c, hence Joule-Thomson effect can be applied for cooling effectively. At the inversion temperature, no rise or fall in temperature of a gas occurs while expanding. But above the inversion temperature, the gas gets heated up when allowed to expand through a hole.

There are different methods used for liquefaction of gases:

1) In **Linde's method,** Joule-Thomson effect is used to get liquid air or any other gas.

2) In **Claude's process,** the gas is allowed to perform mechanical work in addition to Joule-Thomson effect so that more cooling is produced.

3) In **Adiabatic process,** cooling is produced by removing the magnetic property of magnetic material such as gadolinium sulphate. By this method, a temperature of 10^{-4} K i.e. as low as 0 K can be achieved.

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